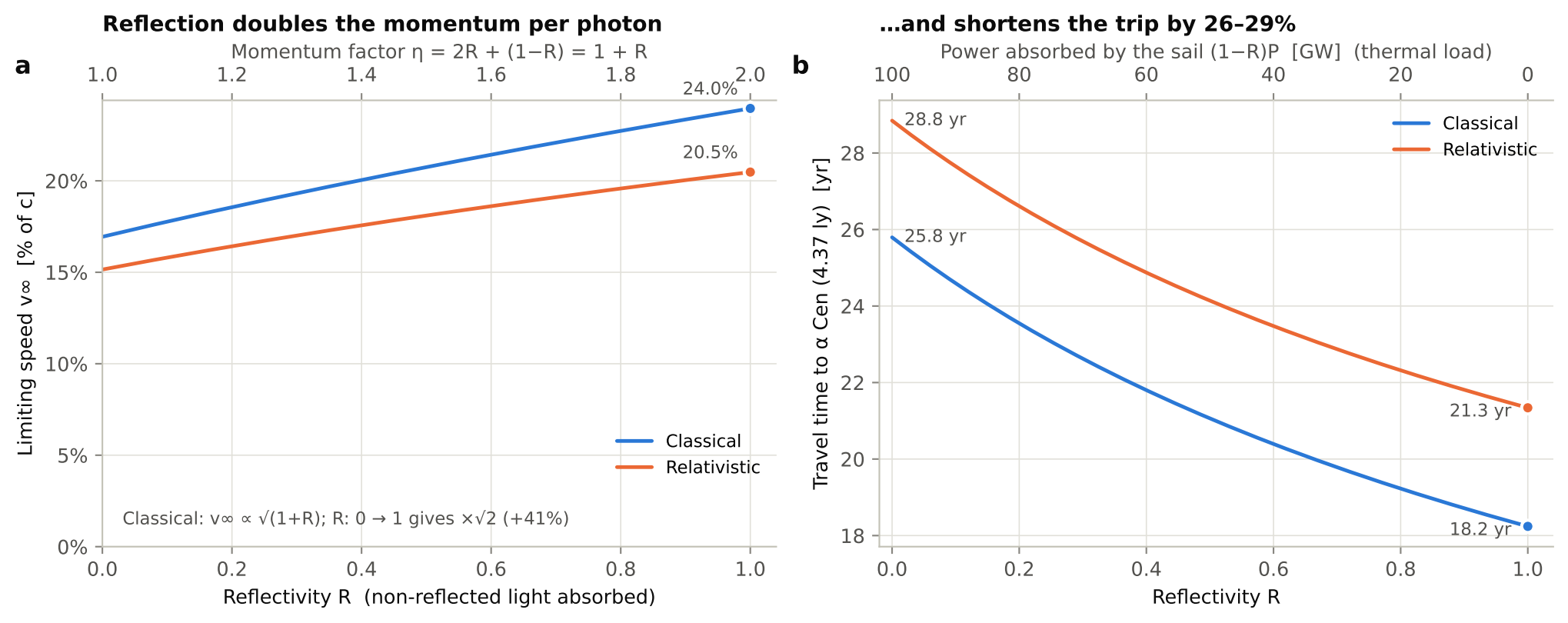
Reflection doubles the momentum per photon
a
Momentum factor η = 2R + (1−R) = 1 + R
1.0
1.2
1.4
1.6
1.8
2.0
24.0%
20.5%
20%
15%
10%
5%
0%
Limiting speed v∞ [% of c]
Classical
Relativistic
Classical: v∞ ∝ √(1+R); R: 0 → 1 gives ×√2 (+41%)
0.0
0.2
0.4
0.6
0.8
1.0
Reflectivity R (non-reflected light absorbed)
…and shortens the trip by 26–29%
b
Power absorbed by the sail (1−R)P [GW] (thermal load)
100
80
60
40
20
0
28.8 yr
25.8 yr
21.3 yr
18.2 yr
28
26
24
22
20
18
Travel time to α Cen (4.37 ly) [yr]
Classical
Relativistic
0.0
0.2
0.4
0.6
0.8
1.0
Reflectivity R